



Financial news sentiment meets market data: A large language model-based approach to stock price prediction

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HIGHLIGHTS

- We propose a framework integrating LLM sentiment analysis into stock forecasting.
- We compared 3 LLM sentiment extractors and 3 stock forecasting models.
- In our analysis LSTM achieves the best trade-off between accuracy and efficiency.
- Vicuna achieves the best sentiment extraction despite higher hallucination rates.

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ABSTRACT

In this paper, we present a stock market forecasting framework that integrates sentiment analysis of news headlines associated with individual stocks performed by Large Language Models (LLMs) with economic indicators. Specifically, we infer multi-class labels and continuous polarity scores by using LLMs (i.e., Llama, Vicuna, and Mistral) under zero-shot settings from news content. These sentiment signals are combined with historical price data and economic indicators and fed as input to different deep learning models (i.e., Long Short-Term Memory (LSTM) networks, Generative Adversarial Networks (GANs), and Transformers) for predicting stock closing prices. We evaluate the proposed framework on a dataset comprising historical financial time series and 8652 news articles related to 47 meme stocks listed on major stock exchanges, covering the period from January 2019 to December 2021. While Vicuna delivers the quickest sentiment processing (around 14% faster than Llama), it achieves the greatest hallucination rate (9.6%). Among the forecasting models, Transformer architectures offer improved Mean Absolute Percentage Error (MAPE) scores (9.995–12.049%) while LSTM requires minimal computational resources for training (up to 6%).

1. Introduction

The stock market serves as a relevant indicator of economic performance, representing national economies with its dynamics and returns, making stock forecasting one of the most challenging tasks [1]. Traditional forecasting approaches have relied on the analysis of technical indicators, macroeconomic factors, and company-specific data [2]. With the growing availability of heterogeneous data sources, the focus has shifted toward integrating numerical time series with unstructured textual information [3]. However, effectively

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combining these modalities to capture temporal and contextual dependencies in a unified predictive framework remains a significant challenge, particularly in achieving robust and adaptive stock market forecasting models [4].

In recent years, there has been increasing academic recognition of the critical role investor sentiment plays in shaping financial markets [5], although it is significantly influenced by several contextual factors (e.g., news events, brand strength, and loyalty). Sentiment analysis further contributes to sustained investor interest and can impact stock performance [6].

While the traditional text mining methodologies have shown some effectiveness in analyzing content from online stock forums, they typically depend on labor-intensive preprocessing and the availability of large, labeled datasets [7]. These requirements limit their scalability and hinder their capacity to reflect the real-time and complex dynamics of investor sentiment. In turn, large language models (LLMs) such as ChatGPT¹ and LLaMA,² which leverage in-context learning, offer a more flexible and efficient alternative. These models can extract sentiment with greater nuance from large volumes of financial textual content, eliminating the need for task-specific supervision and enabling more accurate and timely sentiment-driven analysis of stock performance [7].

Financial time series forecasting plays a crucial role in quantitative finance, providing valuable insights that drive the development of effective and profitable investment strategies [8]. However, stock price prediction remains a challenging task due to the nonlinear and nonstationary nature of financial time series, which are shaped by a multitude of dynamic factors [1]. Indeed, price fluctuations are the result of complex dynamics among macroeconomic policies, societal fundamentals, and investor sentiment [9], contributing to a highly volatile and uncertain market environment.

1.1. Gaps

Stock prices are often modeled as univariate time series data, which form the foundation of most forecasting models [10]. Although these data are continuously recorded, they are often noisy and susceptible to anomalies due to the influence of external events [11]. Additionally, traditional models tend to struggle with generalization, particularly when applied to unseen or out-of-distribution data. This is largely because one-dimensional time series data offer relatively simplistic features, primarily capturing short-term price fluctuations without sufficient contextual representation [12]. However, recent advances in computing resources and data processing have enabled deep learning models to overcome some of these limitations by learning richer, hierarchical representations from raw time series data, thereby enhancing robustness and predictive accuracy [11].

Despite the progress achieved by deep learning, challenges surrounding universality and interpretability remain central concerns in stock market forecasting. Many existing neural network-based models exhibit limited generalizability, often performing well only on specific datasets or under narrow market conditions [13]. However, as economic globalization deepens, financial markets are no longer isolated systems: they are complex, interdependent networks involving diverse countries, sectors, and asset classes. This evolving landscape underscores the urgent need for models that are both broadly applicable across varying market contexts and capable of providing interpretable predictions. Addressing these issues is a key direction for future research in stock forecasting, especially as the demand grows for AI systems that not only perform well but also offer transparent and trustworthy decision-making processes.

Most existing research in stock market time series forecasting has concentrated on short-term predictions, typically forecasting prices on single indices (e.g., [14]).

Beyond these architectural and scope limitations, the integration of LLMs into financial forecasting itself remains hindered by four concrete unresolved challenges that, taken together, motivate the present study.

First, we can take into account the limitations of domain-specific LLMs. Indeed, open-source financial LLMs (e.g., FinBERT, FinGPT, InvestLM) are typically smaller models, often trained primarily on formal news content, which limits their generalizability across diverse financial NLP tasks and unstructured textual environments. Larger proprietary models such as BloombergGPT are not publicly available, preventing independent assessment by the research community. This motivates the evaluation of general-purpose, publicly available LLMs (Llama, Mistral, Vicuna) as zero-shot sentiment extractors, which is a topic that has been underexplored.

Another thing to be taken into account is the non-trivial problem of identifying the optimal prompt for sentiment extraction in a domain characterized by uncertainty, volatility, and non-linear behavior. Different LLMs may respond differently to the same prompt, and the literature lacks a systematic comparison under identical prompting conditions, a gap our study directly addresses by using a fixed prompt and varying the LLM.

The third element to be taken into account is the scarcity of high-quality financial-textual datasets, whose building demands rare combined expertise in financial knowledge and textual semantics, limiting the size and diversity of evaluation corpora reported in the literature. Prior LLM-based sentiment-augmented forecasting works typically rely on small textual samples ($\approx [1,000]1000$ headlines), which are insufficient for robust comparative conclusions across architectures. Our study uses [8,652]8652 headlines, providing a substantially larger evaluation base.

Lastly, the lack of identification of effective predictive features also needs to be considered. Indeed, while many works generate large numbers of features from historical financial data, these features alone do not always yield accurate forecasts [15]. Whether and how LLM-derived sentiment features meaningfully complement classical economic indicators across different deep learning architectures (LSTM, GAN, Transformer) is an open empirical question that no existing study addresses systematically.

¹ <https://chatgpt.com>.

² <https://www.llama.com>.

The intersection of these four challenges, and their particular relevance to the underexplored domain of meme stocks [16], whose dynamics are dominated by social-media-driven retail behavior rather than fundamentals, provides the concrete justification for our research question. Specifically, prior works such as [7] evaluate only a single LLM on approximately [1,000]1000 news samples, while [15] rely solely on historical price data and economic indicators without any textual signal.

To address the identified gaps, we combined daily historical prices of 47 companies with sentiment information extracted by LLMs from news headlines and additional economic indicators to predict daily closing prices using deep learning models.

1.2. Research contributions

In this paper, we address the challenge of improving stock price forecasting by integrating sentiment analysis of financial news headlines using LLMs along with complex economic indicators. Our objective is to infer nuanced sentiment signals from unstructured textual data through LLMs, which are then incorporated into deep learning-based predictive models to enhance forecasting effectiveness. Our approach has been focused on meme stocks that are mostly affected by user behavior on social media. The proposed framework has practical implications that can be applied across application domains that combine textual data and time series, such as event detection, e-health diagnosis, and so on.

The main contributions of this work are the following:

- We propose a framework integrating zero-shot LLM-driven sentiment analysis of financial news headlines with historical stock price data, enhancing the accuracy of closing price prediction;
- We integrate derived economic indicators that capture short-term price momentum, trend direction, and multi-scale temporal signals obtained through frequency-based analysis;
- We evaluate the performance in terms of effectiveness and efficiency of three LLMs for polarity extraction and three deep learning architectures within this framework to assess their forecasting capabilities on 47 prominent publicly traded companies across major global stock exchanges.

The remainder of the paper is organized as follows. [Section 2](#) reviews the related work in stock market prediction. [Section 3](#) describes the stock forecasting task. [Section 4](#) outlines the proposed framework for integrating LLM-based sentiment with deep learning models. [Section 5](#) details the experimental design, model configurations, and evaluation metrics. [Section 6](#) presents and discusses the results of our empirical evaluation. Finally, [Section 7](#) concludes the paper and highlights directions for future research.

2. Related work

Different from general time series forecasting tasks [17], stock price prediction is influenced by the cognitive biases, emotional responses, and sociopolitical dynamics of chaotic human behavior. Indeed, the rising availability of unstructured textual data, particularly financial news and social media content, has opened new alternatives for adding sentiment-driven insights into predictive models [18].

Integrating Natural Language Processing (NLP) with financial forecasting has emerged as a crucial research field, encompassing several challenges such as the extraction of meaningful sentiment signals from financial text, the temporal alignment and aggregation of sentiment features with traditional financial indicators, and the development of robust predictive models capable of leveraging multimodal information sources [19].

In this regard, recent advances in language models have changed the landscape of sentiment analysis, offering opportunities for understanding financial context without task-specific fine-tuning. However, the application of these models to financial forecasting implies distinct issues, including handling sparse news coverage, managing temporal relationships in sentiment data, and picking appropriate architectural frameworks for integrating textual and numerical elements [11].

The approaches to stock market prediction can be characterized depending on their methodological focus and data utilization methodologies:

1. technical analysis approaches, which concentrate exclusively on past price and volume data, employing mathematical indicators to detect patterns and trends;
2. fundamental analysis techniques, which use company-specific financial measurements, earnings data, and macroeconomic indicators to assess intrinsic value and predict price movements;
3. quantitative approaches, which apply statistical and machine learning techniques, spanning from conventional econometric models to sophisticated deep learning-based models, to capture intricate nonlinear connections in financial data;
4. sentiment-driven techniques, which integrate textual input from news items, social media, and financial reports to bring market psychology and investor behavior into prediction models;
5. hybrid techniques, which integrate several strategies to harness the complementary properties of distinct data sources and analytical tools.

Recent studies have increasingly focused on relevant textual information to predict stock movement and extract further informative text features [20]. Indeed, analyzing the emotional orientation of the textual context helps in understanding the stock market's emotional context and its dynamic fluctuations more deeply [6]. For instance, Liu et al. [21] propose FinBERT, a pre-trained language model based on BERT, trained on general and financial corpora using multi-task self-supervised learning across six pre-training tasks,

achieving state-of-the-art performance. Effectively leveraging user-generated content in social communities remains a significant challenge for sentiment analysis, particularly due to the presence of low-quality comments that may compromise the performance of approaches that rely on opinion-mining [22].

In recent years, researchers have explored fundamental analysis, sentiment analysis, and related methods, trying to enhance prediction effectiveness. For instance, the combination of technical indicators and text has been employed to improve predictive performance and provide deeper insights into market dynamics [11]. Traditional approaches, such as hard fusion, apply simple, almost parameter-free methods to combine data [23], whereas soft fusion leverages deep learning techniques to extract richer feature representations and enable deeper interaction between modalities [24]. Another framework has been made by Liu et al. [25], which proposes a multimodal emotion recognition framework that learns from LLMs via imitation, enabling training on unlabeled real-world stock market data by generating pseudo labels from textual input.

Whilst the combination of textual and numerical features is receiving much attention, fundamental analysis remains a widely adopted method in finance since valuable information can be extracted from stock prices. Mining the information cloistered in price fluctuations still enables the prediction of the stock market's trend [26].

However, the dynamic and volatile nature of the financial markets poses a major challenge to modeling temporal and non-linear relationships, which are critical for accurate time series forecasting [27]. Indeed, temporal dynamics embody intricate, evolving patterns driven by exogenous market events and shifts in investor sentiment. Nonlinearities stem from intricate interdependencies among financial variables, where even minor fluctuations in one factor may trigger large and unpredictable effects [28]. Stock market prediction is further hindered by the difficulty of distinguishing signal from noise, data biases (e.g., survivorship, look-ahead bias), and overfitting risks that complicate model generalization. These factors limit the reliability of forecasts and the practical deployment of predictive systems in dynamic financial environments.

Another main problem in the finance domain is the feature extraction process, which is often performed manually or semi-automatically by using appropriate rules [26]. Indeed, existing tools for analyzing investor sentiment are constrained by their reliance on large, high-quality labeled datasets, the creation of which is both costly and labor-intensive. A further hurdle in designing predictive models concerns the need for effective representation of temporal patterns in financial time series, which are non-linear, volatile, and dynamic [4]. Additionally, these models struggle to adapt to the changing dynamics of the market and often fail to capture the subtle and context-dependent sentiments embedded in unstructured financial text, which limits their overall performance and practical applicability. This study contributes to the hybrid techniques category by systematically integrating LLM-based sentiment analysis with traditional technical indicators across multiple deep learning architectures, addressing the fundamental challenge of multimodal financial forecasting in an era of abundant textual data.

The main novelties of the proposed approach w.r.t. the state of the art are summarized as follows:

- We integrate LLM-driven sentiment analysis with historical financial time-series data and economic indicators to handle the dynamic behavior of the market, in contrast to prior works such as [15], and [4] which use only historical data and economic indicators;
- We deal with a financial time series forecasting problem while also considering the sentiment extracted by LLMs. In contrast, [7] only studies the correlation between LLM-driven sentiment analysis and time series;
- We assess the performance of three different LLM models used for sentiment analysis with a dataset composed of 8652 news headlines, in contrast to [7] which only uses one LLM and 1000 pre-filtered samples.

3. Task

Stock prediction is typically approached as a time series forecasting problem [4], in which the aim is to forecast future price changes using past data and other inputs.

In our context, we focus on the *time series* forecasting problem (see Definition 1), in which we consider multivariate information (i.e., news, reports, historical data) to estimate future stock prices based on previous data and other relevant variables (i.e., sentiment information). This undertaking is particularly arduous due to the complicated, non-linear, and generally non-stationary structure of financial markets [18].

Definition 1.

We aim to find the stock price P_{t+h} at the instant $t + h$, such that:

$$P_{t+h} = f(P_{(t:t-n)}, X_t^{economic}, X_{(t:t-k)}^{sent}) + \varepsilon \quad (1)$$

where:

- $P_{(t:t-n)} = [P_t, P_{(t-1)}, \dots, P_{(t-n)}]$ represents the historical price sequence over n periods;
- $X_t^{economic}$ represents fundamental data available at time t , such as financial ratios, earnings metrics, and macroeconomic variables;
- $X_{(t:t-k)}^{sent}$ represents sentiment features derived from textual data over a window of k periods;
- $f(\cdot)$ denotes the prediction function that models the relationship between both the past price P_t and the exogenous variables X_t ;
- $\varepsilon_t \sim \mathcal{N}(0, \sigma_t^2)$ represents the error term, allowing for time-varying volatility.

This formulation explicitly takes into account the multimodal nature of financial prediction, incorporating different types of information that may operate on different time scales and exhibit varying degrees of persistence.

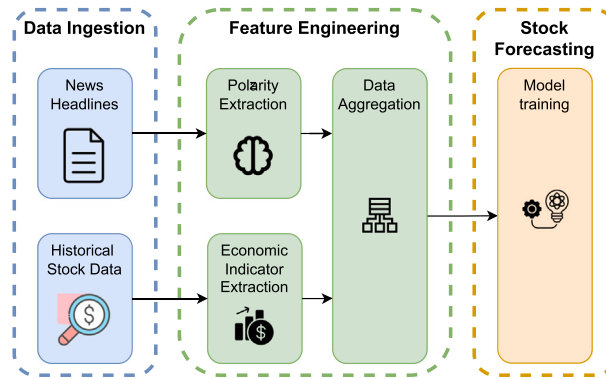


Fig. 1. Overview of the data flow of the proposed framework. The framework consists of three main phases: (i) data ingestion of historical stock prices and news headlines; (ii) feature engineering to extract and aggregate sentiment polarity and economic indicators; (iii) stock forecasting, using the engineered features to train a model to predict stock closing prices.

Table 1

Dataset characterization of the news for the 47 meme stocks.

	Mean	Std	Min	Max
Stats	919.936	209.61	490	1044

4. Methodology

We design a multimodal framework for financial forecasting, integrating traditional economic indicators with sentiment analysis scores inferred from LLMs on financial news to predict stock price returns. As shown in Fig. 1, our framework contains three main steps: data collection, feature engineering, and model training.

In the *data ingestion* phase, we collect information from two different streams: historical financial information and daily news. This information is fed as input to the *feature engineering* phase to infer economic indicators and sentiment polarity of the daily news. We then aggregate data and perform the *training* of a deep learning model to predict future stock values.

4.1. Data ingestion

Both historical price and news headlines data were provided by a private source. The historical price data covered daily open, high, low, close, and volume records. News headlines, instead, provided publication date, source URL, and company ticker for each article.

We ingest data from two different data sources, which include historical, financial, and news-related data for 47 meme stocks listed on major stock exchanges from January 2019 to December 2021. The former is made up of daily records of opening, high, low, and closing prices and daily exchange volumes, while the latter is composed of 8652 headlines, their publication date, a link to the source, and the corresponding company acronym. The included enterprises represent a diverse sample of companies (e.g., technology, transportation), allowing for a realistic and nuanced evaluation of the methodologies proposed in this study.

Table 1 reports the characterization of news for the 47 meme stocks, whose price dynamics are predominantly influenced by social media-driven narratives rather than traditional fundamental indicators.

The prediction of meme stock prices presents different challenges, primarily due to the dynamic nature of social media-driven behaviors and the high volatility of these equities, which are strongly influenced by both financial indicators and contextual factors [16].

The multimodal structure of the dataset enables a comprehensive analysis of possible correlations between sentiment inferred from news textual content and stock price fluctuations.

4.2. Feature engineering

The preprocessing pipeline begins with the systematic loading of economic data for each ticker symbol from serialized pickle files, converting each to a 32-bit floating-point precision. This standardization ensures consistency across securities and improves memory efficiency during downstream computations. Subsequently, the feature engineering step is performed. This step is divided into polarity extraction from news headlines and economic indicator extraction from numerical time series. Once both processes are completed, the results are merged through a data aggregation step, aligning historical financial data with corresponding sentiment information.

```

You are an expert in finance and natural language
processing. Analyze the sentiment of the given text
and provide scores for four categories:
-Positive: Measures the degree of positive sentiment
(0.0 to 1.0);
-Negative: Measures the degree of negative sentiment
(0.0 to 1.0);
-Neutral: Measures the degree of neutral sentiment
(0.0 to 1.0);
-Polarity: Overall sentiment polarity (-1.0 to 1.0,
similar to VADER's compound score).

Rules:
-Positive + Negative + Neutral must sum to 1.0;
-Polarity ranges from -1.0 (extremely negative) to
1.0 (extremely positive);
-Consider financial context and nuances in your
analysis;
-Provide only the JSON output, no additional
explanation.

```

Listing 1. Prompt used for polarity extraction from news headlines.

4.2.1. Polarity extraction

The first step of feature engineering involves the extraction of sentiment elements from each headline. To achieve this, we deploy three different LLMs (i.e., Llama, Mistral, and Vicuna) in a zero-shot setting, relying exclusively on their general-purpose capabilities to analyze and assign sentiment labels to financial news headlines.

We prompted the LLMs to categorize headlines into three unique sentiment categories through the prompt shown in [Listing 1](#): *positive*, reflecting news that anticipates likely increases in stock values, *negative*, indicating news that implies probable downward pressure on stock prices, and *neutral*, representing news with no major anticipated impact on stock price movements. Furthermore, LLMs are instructed to report an overall sentiment polarity score in the range of $[-1.0, 1.0]$, establishing a continuous measure of emotion intensity. These four fields (positive, negative, neutral, and polarity) are schema-enforced in a JSON object. To ensure logical consistency, we validate the parsed results: if the sum of the three category probabilities falls outside the range $[0.99, 1.01]$, the response is flagged as a hallucination. In such cases, the model is re-prompted with the same headline until a valid, normalized distribution is achieved.

To tackle the challenge of the fluctuating influence of news sentiment on stock price movements, we introduce an impact factor that estimates the potential market effect of daily sentiment. This addition overcomes the limitations of raw sentiment scores by incorporating news volume and sentiment consistency. Specifically, the impact factor considers the mean polarity of daily headlines, the volume of news coverage, and the standard deviation of sentiment scores, capturing the intuition that high-volume, polarized news days are likely to have greater market impact than days with few headlines or consensus. To handle periods without news coverage, we apply an exponential decay mechanism with a decay rate of 0.05 to propagate the last known sentiment information forward to empty calendar days, representing the gradual lowering influence of past news over time. Additionally, we assign a neutral sentiment value (polarity and neutral scores set to 1, negative and positive scores set to 0) to days without news and introduce a binary flag to distinguish between observed and imputed sentiment, enabling models to differentiate genuine from inferred signals.

Finally, to incorporate company-specific characteristics, we employed a sophisticated embedding approach using the pre-trained FinBERT [21], which processed each ticker symbol to generate contextual embeddings, allowing us to capture semantic relationships between different companies while maintaining computational tractability.

4.2.2. Economic indicator extraction

The second step of our feature engineering method is the production of technical indicators that capture various aspects of market behavior. Economic indicators based on prices are the foundation of time series analysis, since they can capture the fundamental dynamics of asset price movements, providing information about both the magnitude and direction of price changes over time. Therefore, to represent short-term price momentum, we computed percentage changes ([Eq. 2](#)) to identify relative price movement between consecutive periods and logarithmic returns ([Eq. 3](#)) to ensure that returns are additive over time and roughly distributed in a normal way to make econometric analysis smoother.

$$\text{pct_change}_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (2)$$

$$\text{log_change}_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (3)$$

where P_t is the closing price at time t .

We also included a 7-day exponential moving average (EMA) as defined in [Eq. \(4\)](#), the 7-day relative strength index (RSI) as defined in [Eq. \(5\)](#), and the raw stochastic value (RSV) as defined in [Eq. \(6\)](#) to identify and quantify the direction and strength of price

trends, providing important information preparatory to understanding market oscillations.

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1} \quad (4)$$

where $\alpha = \frac{2}{n+1}$ is the smoothing factor and n is the number of days we consider the price P_t ;

$$RSI_t = 100 - \frac{100}{1 + RS} \quad (5)$$

where

$$RS = \frac{\text{Average Gain over 7 periods}}{\text{Average Loss over 7 periods}}$$

$$RSV_t = \frac{C_t - L_n}{H_n - L_n} \times 100 \quad (6)$$

where C_t is the current closing price, L_n is the lowest price over n periods and H_n is the highest price over n periods, with $n = 7$ days.

To quantify the degree of price variation and market uncertainty, essential elements for risk assessment and trading strategy, we also evaluated the average true range (ATR) indicator with a 7-day window to capture recent volatility patterns (Eq. 7) and the Bollinger bands (Eq. 8) calculated over two timeframes which allowed the model to consider both short-term and medium-term volatility regimes. Additionally, we computed the moving average convergence divergence (MACD) indicator, which provided insights into momentum shifts and potential trend reversals (Eq. 9), unfolding changes in strength, direction, momentum, and duration of price.

$$ATR_t = \frac{1}{n} \sum_{i=0}^n TR_i \quad (7)$$

where True Range (TR) is defined as

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

with H_t , L_t , and C_t representing high, low, and closing prices, respectively, and $n = 7$ days;

$$\begin{aligned} \text{Upper Band}_t &= SMA_t + k \cdot \sigma_t \\ \text{Lower Band}_t &= SMA_t - k \cdot \sigma_t \end{aligned} \quad (8)$$

where SMA_t is the simple moving average, σ_t is the standard deviation over the period, and k is typically 2, which defines the bands' amplitude.

$$MACD_t = EMA_{short}(P_t) - EMA_{long}(P_t) \quad (9)$$

where the EMA_{short} uses a 3-day period and the EMA_{long} uses an 11-day period, with a 7-day signal line.

Finally, we also perform frequency domain analysis through Fourier transform decomposition of closing prices, an approach that filters out high-frequency noise while preserving the underlying structural patterns in price movements, creating features that capture different scales of market dynamics. To prevent data leakage, the Fast Fourier Transform (FFT) decomposition is computed exclusively on the training portion of each ticker's closing price series; the resulting frequency components are then applied consistently to all splits.

For this purpose, the closing price series was decomposed using the FFT as defined in Eq. (10).

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j \frac{2\pi}{N} kn}, \quad k = 0, 1, \dots, N-1 \quad (10)$$

where $x(n)$ represents the closing price series. We then reconstruct the signal considering only signals that appear every 3, 9, and 15 days.

These three reconstruction periods were selected to capture short-term (approximately intra-week), medium-term (approximately bi-weekly), and longer-term (approximately monthly) market cycles.

4.2.3. Data aggregation

The last step of the feature engineering is the data aggregation, in which we merge and align the two indicators extracted during the previous phases. We reindexed all data to a daily frequency, maintaining constant temporal spacing among stocks and omitting weekend observations to focus entirely on active trading hours. The aggregation process consolidates multiple headlines per day into the mean, median, and standard deviation of sentiment scores across all three LLMs, making it possible to capture both the central tendency and variability of daily sentiment, providing a comprehensive representation of the news environment. As a final step, this dataset was merged on a daily basis with the economic one, enriched with the economic features discussed in the previous subsection.

4.3. Stock forecasting

In this section, we first present the proposed data-driven module for stock market forecasting, relying on recent state-of-the-art approaches according to the classification defined by Bao et al. [18], and, successively, we describe the training procedure in Section 4.3.1.

LSTM architecture represents a hybrid neural network design that integrates learnable stock embeddings with bidirectional temporal processing for multi-asset financial forecasting, addressing the dual challenges of capturing company-specific characteristics and temporal dependencies inherent in financial time series predictions.

GAN architecture employs a dual-network design constituted by a residual generator and discriminator, both containing, as in this case, learnable stock embeddings. This architecture tries to exploit adversarial training dynamics to capture the intricate patterns and correlations inherent in multi-asset financial data.

Transformers architecture is built on the attention neural network design that employs self-attention mechanisms and positional encoding to capture temporal correlations in financial time series data without relying on recurrent components. This model mixes learnable stock embeddings with multihead self-attention to handle the hard task of simulating long-range interdependence while preserving company-specific traits. The positional encoding mechanism introduces temporal order information into the embedded representations, generating, in this way, position-dependent signals through the application of sine functions to even-indexed dimensions and cosine functions to odd-indexed ones, creating unique positional signatures for each temporal position within the sequence.

4.3.1. Training

The training phase uses a thorough technique designed to provide robust model evaluation and optimal performance across multiple architectural configurations.

Feature preprocessing comprised a normalization within a range of [1, 6] to ensure numerical stability and to avoid both vanishing gradient problems and collapsing around zero values. Normalization is performed independently for each ticker symbol to account for differences in price scales and volatility across stocks. A sliding window approach is used to generate sequential input–output pairs with a window length of five days. However, to ensure a fair evaluation, the dataset was partitioned into training (70%), validation (20%), and test (10%) sets.

We train the model following Algorithm 1. The core training process utilized a loss function with L1 regression loss, integrating an L1 regularization term on linear layer weights to prevent overfitting and maximum error weighting to penalize extreme prediction errors that could indicate model instability. This multi-component method guaranteed that models addressed the dual nature of financial forecasting, where both price levels and movement directions hold important practical value. During the training, we employed the approach of gradient trimming to prevent bursting gradients, a frequent issue in recurrent architectures when processing large sequences. On the validation dataset, after each training epoch, we evaluated model performance, implementing model checkpointing based on the decrease in the loss function to preserve only configurations that achieved improved validation performance, as an early stopping mechanism to prevent overfitting, ensuring optimal model selection.

Multiple performance indicators were simultaneously monitored (i.e., MAE, MSE, RMSE, and \$MAPE) alongside percentage errors, a financial-specific measure, other than computational efficiency metrics related to training and inference time, which provided insights into the practical deployment viability of different model configurations. AdaGrad was chosen as the optimization algorithm with an adaptive learning rate schedule based on monitoring the average loss, which automatically reduced learning rates when validation performance plateaued for consecutive epochs, allowing aggressive initial learning but ensuring convergence stability in later training phases. We also structured a weight decay regularization offering extra overfitting protection, which is extremely significant given the complicated feature interactions present in the sentiment-augmented datasets, where high-dimensional feature spaces could lead to spurious correlations.

At the end of the training phase, we conducted a final evaluation using a completely isolated test dataset that had remained unseen throughout the training and validation phases. We stored all test metric results as final accuracy model results.

5. Experimental setup

We trained and compared three model architectures during the stock forecasting phase: LSTM, GAN, and a Transformer-based model. Each architecture was tested under four feature engineering configurations. The first serves as a baseline using only economic indicators, while the remaining three incorporate polarity information extracted from news headlines using distinct open-source LLMs. This setup allows us to assess how differences in sentiment extraction across models influence forecasting performance. We used the following LLMs in their quantized Q_{4K_M} variants to balance inference performance with computational efficiency, enabling the polarity extraction to be executed on a local machine: i) Llama-3.1-8B-Instruct,³ ii) Vicuna-7B-v1.5,⁴ and iii) Mistral-7B-Instruct-v0.3.⁵

³ <https://huggingface.co/bartowski/Meta-Llama-3.1-8B-Instruct-GGUF>.

⁴ <https://huggingface.co/TheBloke/vicuna-7B-v1.5-GGUF>.

⁵ <https://huggingface.co/bartowski/Mistral-7B-Instruct-v0.3-GGUF>.

Algorithm 1 General steps and training procedure followed.**Require:** Dataset D , Window size $w=5$ **Ensure:** Trained model $\mathcal{M}$, Metrics $\mathcal{R}$

```

1: Split  $D$  into train (70%), val (20%), test (10%)
2: Fit scaler on train set only
3: for all ticker  $t$  in  $D$  do
4:   Normalize  $t$  using the fitted scaler
5: end for
6: Create sliding windows of length  $w$  for seq2seq inputs
7: Init model  $\mathcal{M}$  with ReLU activations
8: Define  $\mathcal{L} = \text{L1Loss} + \lambda_1 \cdot \text{L1Reg} + \lambda_2 \cdot \text{MaxErrPenalty}$ 
9: Init AdaGrad optimizer with adaptive LR
10: Set gradient clipping threshold  $\theta$ 
11: for each epoch do
12:   for mini-batch in train set do
13:     Forward  $\mathcal{M}$ ; compute  $\mathcal{L}$ ; clip gradients; backprop
14:   end for
15:   Evaluate on val set; update best model if improved
16:   if early stopping triggered then
17:     break
18:   end if
19: end for
20: Evaluate  $\mathcal{M}$  on test set
21: Compute metrics: MAE, MSE, RMSE, $MAPE, %Error, runtime
22: return  $\mathcal{M}, \mathcal{R}$ 

```

Table 2
Model summary of LSTM Model.

Layer	Output Shape	Param #
LSTMModel	[32, 1]	123,008
Embedding	[32, 5, 120]	5640
LSTM	[32, 5, 192]	635,136
ReLU	[32, 192]	–
LayerNorm	[32, 192]	384
Linear	[32, 128]	24,704
SELU	[32, 128]	–
Linear	[32, 64]	8256
LayerNorm2	[32, 64]	128
Linear	[32, 1]	65
Total parameters		797,321
Trainable params		797,321
Non-trainable params		0

5.1. Stock forecasting models: implementation details

LSTM architecture. Looking at the general architecture in Table 2, we concatenate an embedding layer that maps discrete stock identifiers to dense vector representations with input features to create enriched representations that combine quantitative market data with learned stock-specific characteristics. First, we used an orthogonal technique to initialize the weights of the LSTM to maintain gradient stability and prevent vanishing gradients and the initialization with appropriate scaling factors for linear layers.

Moreover, the core temporal processing utilizes a bidirectional LSTM for superior pattern recognition that processes sequences capturing comprehensive temporal context within each sequence window, applying dropout regularization between LSTM layers when the network depth exceeds one layer to balance model capacity with overfitting prevention. Layer normalization with learnable affine parameters stabilizes training dynamics, while a combination of SELU activation provides self-normalizing characteristics and ReLU activation prevents saturation.

The final prediction layer maps processed representations to target output dimensions.

GAN architecture. The fundamental element for both the generator (Table 3) and the discriminator (Table 4) is the residual block. It implements a two-layer convolutional structure and a normalization layer applied before the first convolution operation. To ensure computational efficiency and reliable signal propagation, both a ReLU activation function positioned between consecutive layers and post-processing normalization were incorporated in the residual block. Skip connections allow dimensional mismatches by 1×1

Table 3
Generator model summary.

Layer	Output Shape	Param #
Generator	[32, 1]	–
Embedding: 1–1	[32, 5, 120]	5640
GRU: 1–2	[32, 5, 512]	1,018,368
Dropout: 1–3	[32, 5, 512]	–
ResidualBlock: 1–4	[32, 512, 5]	1024
LayerNorm: 2–1	[32, 512, 5]	5120
Conv1d: 2–2	[32, 256, 5]	393,472
ReLU: 2–3	[32, 256, 5]	–
Conv1d: 2–4	[32, 512, 5]	393,728
LayerNorm: 2–5	[32, 512, 5]	5120
Conv1d: 2–6	[32, 512, 5]	262,656
Conv1d: 1–5	[32, 256, 5]	131,328
ResidualBlock: 1–6	[32, 256, 5]	512
LayerNorm: 2–7	[32, 256, 5]	2560
Conv1d: 2–8	[32, 128, 5]	98,432
ReLU: 2–9	[32, 128, 5]	–
Conv1d: 2–10	[32, 256, 5]	98,560
LayerNorm: 2–11	[32, 256, 5]	2560
Conv1d: 2–12	[32, 256, 5]	65,792
Conv1d: 1–7	[32, 128, 5]	32,896
ResidualBlock: 1–8	[32, 128, 5]	256
LayerNorm: 2–13	[32, 128, 5]	1280
Conv1d: 2–14	[32, 64, 5]	24,640
ReLU: 2–15	[32, 64, 5]	–
Conv1d: 2–16	[32, 128, 5]	24,704
LayerNorm: 2–17	[32, 128, 5]	1280
Conv1d: 2–18	[32, 128, 5]	16,512
Conv1d: 1–9	[32, 64, 5]	8256
LayerNorm: 1–10	[32, 5, 64]	128
Linear: 1–11	[32, 5, 32]	2080
Dropout: 1–12	[32, 5, 32]	–
Linear: 1–13	[32, 5, 16]	528
LayerNorm: 1–14	[32, 5, 16]	32
Linear: 1–15	[32, 5, 1]	17
LeakyReLU: 1–16	[32, 5]	–
Linear: 1–17	[32, 1]	6
Total params		2,597,487
Trainable params		2,597,487
Non-trainable params		0

convolution operations if input and output channel dimensions differ, guaranteeing adequate gradient flow maintenance across the whole network depth and preventing degradation.

Both the generator and discriminator commence processing using stock embeddings that are concatenated with input features and then concatenated with output features.

The generator adopts a hybrid temporal processing methodology that blends recurrent and convolutional components to achieve full sequence modeling capabilities.

The discriminator employs a convolutional feature extraction workflow through three residual blocks that demonstrate increasing channel complexity, aimed at capturing temporal patterns and correlations within the financial sequences. To produce fixed-size representations that remain independent of input sequence length and reduce the temporal dimension, an adaptive average-pooling layer was utilized following the feature extraction pipeline. This output was concatenated with the stock embedding, and the resulting tensor was passed through a normalization and linear transformation layer implementing a dimensional reduction pathway with a ReLU activation function to prevent gradient saturation while maintaining discriminative capability throughout the training process.

Looking at the whole architecture, the training framework operates through alternating optimization between generator and discriminator networks with the generator learning to produce increasingly realistic financial sequences while the discriminator simultaneously develops enhanced capability to distinguish between authentic and synthetic data. Among other things, the embedding integration approach enables both networks to maintain company-specific characteristics throughout the adversarial learning process, ensuring that generated sequences preserve distinct stock qualities.

Transformers architecture. As shown in Table 5, the fundamental computational component incorporates several encoder modules, each of which presents multi-head self-attention mechanisms that make it possible to detect significant temporal correlations without the sequential processing constraints inherent in recurrent designs. The feed-forward section of the encoder block incorporates a two-layer architecture with ReLU activation and intermediate dimensional expansion, giving the nonlinear transformation capabilities necessary to capture complicated patterns in financial data. Layer normalization, modeled before both the attention and

Table 4
Discriminator model summary.

Layer	Output Shape	Param #
Discriminator	[32, 1]	–
ResidualBlock: 1–1	[32, 32, 6]	64
LayerNorm: 2–1	[32, 1, 6]	12
Conv1d: 2–2	[32, 32, 6]	128
ReLU: 2–3	[32, 32, 6]	–
Conv1d: 2–4	[32, 32, 6]	3104
LayerNorm: 2–5	[32, 32, 6]	384
Conv1d: 2–6	[32, 32, 6]	64
ResidualBlock: 1–2	[32, 64, 6]	128
LayerNorm: 2–7	[32, 32, 6]	384
Conv1d: 2–8	[32, 64, 6]	6208
ReLU: 2–9	[32, 64, 6]	–
Conv1d: 2–10	[32, 64, 6]	12,352
LayerNorm: 2–11	[32, 64, 6]	768
Conv1d: 2–12	[32, 64, 6]	2112
ResidualBlock: 1–3	[32, 128, 6]	256
LayerNorm: 2–13	[32, 64, 6]	768
Conv1d: 2–14	[32, 128, 6]	24,704
ReLU: 2–15	[32, 128, 6]	–
Conv1d: 2–16	[32, 128, 6]	49,280
LayerNorm: 2–17	[32, 128, 6]	1536
Conv1d: 2–18	[32, 128, 6]	8320
AdaptiveAvgPool1d: 1–4	[32, 128, 1]	–
Embedding: 1–5	[32, 120]	5640
LayerNorm: 1–6	[32, 248]	496
Linear: 1–7	[32, 100]	24,900
LeakyReLU: 1–8	[32, 100]	–
Linear: 1–9	[32, 1]	101
Total params		141,709
Trainable params		141,709
Non-trainable params		0

feed-forward components, implements its pre-normalization variation that has exhibited improved training stability compared to post-normalization alternatives [29]. To enable effective gradient propagation through the network depth while eliminating degradation concerns associated with very deep topologies, residual connections surround both the attention and feed-forward components.

A concatenation of an embedding and a projection layer is transformed to match the model dimension required. The architecture implements a causal attention mask that prevents the model from utilizing future time steps during training, ensuring that predictions rely on historical information and maintain the temporal causality constraints. The last prediction layer pulls the temporal position from the encoded sequence representation and performs a linear transformation to obtain single-step-ahead forecasts. Before transferring the output to this last transformation, to ensure stable output distributions and prevent any scaling concerns, the output itself was processed through a normalization layer.

5.2. Evaluation measure

In this section, we describe the quantitative measurement criteria aimed at examining both traditional regression accuracy and financial-specific forecasting capabilities across all model architectures. For the evaluation of the effectiveness of our framework, we employed classical regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) defined in Eqs. (12) and (13), respectively.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (12)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (13)$$

Alongside standard regression metrics, we employed \$MAPE\$, a variant of the Mean Absolute Percentage Error (MAPE), which provides a scale-independent assessment of prediction accuracy. The implementation, described in Definition 2, addresses the issue of near-zero values using threshold filtering and capping severe percentage errors to prevent a small number of extreme outliers (e.g., predictions on low-price penny stocks during high-volatility events) from dominating the aggregate metric, providing a more

Table 5
Transformer model summary.

Layer	Output Shape	Param #
Transformer model	[32, 1]	–
Linear: 1–1	[32, 5, 256]	7680
Embedding: 1–2	[32, 5, 120]	5640
PositionalEncoding: 1–3	[5, 32, 376]	–
ModuleList: 1–4	–	–
EncoderBlock: 2–1	[5, 32, 376]	–
LayerNorm: 3–1	[5, 32, 376]	752
MultiheadAttention: 3–2	[5, 32, 376]	567,008
Dropout: 3–3	[5, 32, 376]	–
LayerNorm: 3–4	[5, 32, 376]	752
Sequential: 3–5	[5, 32, 376]	96,760
Dropout: 3–6	[5, 32, 376]	–
EncoderBlock: 2–2	[5, 32, 376]	–
LayerNorm: 3–7	[5, 32, 376]	752
MultiheadAttention: 3–8	[5, 32, 376]	567,008
Dropout: 3–9	[5, 32, 376]	–
LayerNorm: 3–10	[5, 32, 376]	752
Sequential: 3–11	[5, 32, 376]	96,760
Dropout: 3–12	[5, 32, 376]	–
EncoderBlock: 2–3	[5, 32, 376]	–
LayerNorm: 3–13	[5, 32, 376]	752
MultiheadAttention:	[5, 32, 376]	567,008
3–14		
Dropout: 3–15	[5, 32, 376]	–
LayerNorm: 3–16	[5, 32, 376]	752
Sequential: 3–17	[5, 32, 376]	96,760
Dropout: 3–18	[5, 32, 376]	–
LayerNorm: 1–5	[5, 32, 376]	752
Linear: 1–6	[32, 1]	377
Total params		2,010,265
Trainable params		2,010,265
Non-trainable params		0

robust view of typical forecasting accuracy. Specifically, the metric sets a minimum threshold of 0.01 for denominator values and caps maximum errors at 100%, ensuring robust statistical interpretation across varying market conditions.

Definition 2. $MAPE = \frac{100\%}{|V|} \sum_{i \in V} \min \left(\left| \frac{y_i - \hat{y}_i}{|y_i| + \epsilon} \right|, C \right)$

where:

- $|V| = \{i : |y_i| > \tau\}$ represents the set of valid indices where actual values exceed the threshold τ ;
- $|V|$ is the cardinality of the valid set;
- ϵ represents the numerical stability constant;
- C is the maximum cap parameter;
- y_i and $\hat{y}_i$ are the actual and predicted values of the stock price.

The evaluation framework integrates an additional percentage error analysis using multiple complementary metrics like *mean percentage error*, *maximum percentage error*, and *minimum percentage error*. These measures provide extensive insights into the distribution and features of prediction errors, enabling the identification of extreme error circumstances that can suggest model instability.

Moreover, to evaluate the computational efficiency of the models, we monitored both *training* and *inference* time provided, where the first one represents the cumulative duration across all epochs for model optimization, while the latter measures the computational cost for processing the test dataset, providing estimates for real-time prediction scenarios.

Finally, we assess the operational overhead by calculating the hallucination rate, which measures the proportion of headlines that require a retry to yield a valid output. High rates not only drain resources but can significantly hinder the scalability and reliability of real-world deployments.

5.3. Hardware and software stack

All studies were conducted on a 2021 MacBook Pro with 1 TB SSD, 10 CPU cores, and 32 GPU cores. The developed Python code included PyTorch for implementing, training, and testing the deep learning models, Pandas and Scikit-learn for data manipulation and preprocessing, and Matplotlib and Seaborn as visualization tools.

5.4. Information for reproducibility

To enable independent reproduction of the methodology, this subsection consolidates all the information required to rebuild an equivalent corpus and replicate every experimental step end-to-end.

Table 6
Generation hyperparameters.

Parameter	Value
seed	42
n_ctx	1024
n_batch	50
temperature	0.7
max_tokens	256
n_gpu_layers	−1
chat_format	chatml

Table 7
Training hyperparameters.

Parameter	Value
Batch size	32
Maximum epochs	200
Early stopping patience	15
Gradient clipping threshold θ	1.0
Loss	$\mathcal{L} = \text{L1Loss} + \lambda_1 \cdot \text{L1Reg} + \lambda_2$
λ_1	10^{-4}
λ_2	0.1
Optimizer	AdaGrad
Learning rate	10^{-2}
Learning rate scheduling η	0.5
Learning rate patience	5 epochs

Data specification. Period: January 2019 – December 2021. Dataset: 47 meme stocks listed on major exchanges. Financial fields collected: opening, high, low, and closing prices and daily volume. News fields collected: headline text, publication date, source URL, and ticker symbol ([8, 652]8652 headlines in total).

LLM inference and polarity extraction. Models used: Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Vicuna-7B-v1.5, all in their Q_{4K_M} quantised variants. Inference framework: llama.cpp for local CPU/GPU execution. Generation hyperparameters are shown in Table 6.

Output enforced via response_format with an explicit JSON schema requiring fields positive, negative, neutral, and polarity. Hallucination handling: when positive + negative + neutral $\notin [0.99, 1.01]$, the model is automatically re-run on the same headline; the retry always produces a valid distribution. The full prompt template is reported in Listing 1.

Feature engineering. Economic indicators are computed according to Eqs. (2)–(9). FTT reconstruction periods: 3, 9, and 15 days, computed on the training portion only. Daily sentiment aggregation: mean, median, and standard deviation of headline-level scores. Missing-news handling: exponential decay of the last known sentiment with rate 0.05, plus a binary imputed-signal flag.

Training protocol. Full per-layer specifications, including output shapes and parameter counts, are reported in Tables 2 for LSTM, Table 3 for GAN generator, Table 4 for GAN discriminator, and Table 5 for the transformer. Sliding-window length $w = 5$. Data split: 70% train / 20% validation / 10% test with the scaler fitted exclusively on the training set and applied (without re-fitting) to validation and test (normalization range [1, 6]). LSTM weights are initialized orthogonally; linear layers use scaled initialization. In Table 7, we report the training hyperparameters.

Evaluation. Exact definitions of MSE, MAE, RMSE, and the custom $\$MAPE$ (with threshold $\tau = 0.01$ and cap $C = 100\%$) are given in Eqs. (11)–(13) and Definition 2. Computational efficiency is reported as cumulative training time and per-test-set inference time.

6. Results and discussion

A comparison of the evaluation of the outcomes on test metrics between the three deep learning architectures reveals major differences in prediction accuracy, computational efficiency, and sentiment integration effectiveness.

The comparative analysis of the three LLMs in Table 8 has revealed distinct performance characteristics in terms of computational efficiency and output reliability for polarity extraction.

From a computational efficiency standpoint, Vicuna is the most efficient model, exhibiting an average generation time of 1.065 seconds per headline, closely followed by Mistral. Llama demonstrates roughly 14% greater computing overhead compared to the most efficient inference model. The alignment of mean and median values across all models indicates steady processing durations with no major outliers impacting the central tendency estimates of generating time. Nonetheless, although Llama necessitates increased computational time, it exhibits remarkable stability with the optimal standard deviation value, signifying highly predictable and consistent processing behavior across various headlines. Both Mistral and Vicuna demonstrate far greater variability in processing durations, indicating that their computational demands vary drastically based on the complexity or attributes of particular headlines.

Table 8

LLM generation metrics. Average generation time with standard deviation, median generation time, and hallucination rate.

Model	Avg generation time	Median generation time	Hallucination rate
Llama	1.216 ± 0.006	1.165	0.0
Mistral	1.092 ± 0.069	1.056	0.048
Vicuna	1.065 ± 0.066	1.019	0.096

Table 9

Forecasting results varying model architecture and LLM used.

Architecture	LLM	loss	MAE	MSE	RMSE	\$MAPE	mean_pct_error	max_pct_error	min_pct_error
LSTM	Llama	3.5880	0.1476	0.3280	0.1850	12.6330	52.7500	108.4100	0.6250
LSTM	Vicuna	3.5070	0.1301	0.3260	0.1630	12.9040	50.0050	105.7220	0.5490
LSTM	Mistral	3.5480	0.1468	0.3390	0.1840	13.4970	51.7570	106.5760	0.5820
LSTM	Economic	3.5210	0.1253	0.3110	0.1570	12.2980	51.9860	106.5560	0.5800
GAN	Llama	4.5620	0.3974	0.3780	0.4980	20.8190	51.6640	129.4850	0.2390
GAN	Vicuna	4.4610	0.3830	0.4050	0.4800	17.8150	52.9500	126.4600	0.2260
GAN	Mistral	4.5040	0.4333	0.5250	0.5430	20.1110	53.0960	126.3610	0.2670
GAN	Economic	4.7180	0.6137	1.2760	0.7690	24.6260	51.5160	126.5240	0.2100
Transformer	Llama	4.4010	0.1444	0.3340	0.1810	12.0490	50.6240	134.7230	0.2680
Transformer	Vicuna	4.2580	0.1436	0.3250	0.1800	11.5040	49.7400	130.0650	0.2960
Transformer	Mistral	4.3260	0.1460	0.3330	0.1830	11.9660	50.3850	132.0340	0.3260
Transformer	Economic	4.2130	0.1317	0.3050	0.1650	9.9950	51.3300	129.0520	0.3410

The most important distinction is related to the rates of hallucination. Indeed, while Llama demonstrates remarkable reliability with no occurrences of invalid responses, consistently delivering accurate sentiment classifications for all analyzed headlines, both Mistral and Vicuna exhibit a moderate hallucination rate with a significantly higher ratio for the latter model, attributable to incorrect formatting or erroneous sentiment extraction.

In Table 9, we show stock forecasting results of each architecture associated with each LLM. LSTM demonstrates exceptional performance across all metrics, reaching the lowest loss values between 3.507 and 3.588 and significantly lower RMSE scores ranging from 0.157 to 0.185. These results indicate a notable enhancement, especially in contrast to the GAN alternative architecture. LSTM models get roughly 60 – 70% lower RMSE while sustaining competitive \$MAPE performance within the 12.298 – 13.497% range. On the other side, transformer architectures offer equivalent RMSE performance with values between 0.165 and 0.183 while achieving marginally superior MAPE scores ranging from 9.995% to 12.049%.

Conversely, GAN models present significant difficulties in attaining accurate magnitude predictions for financial time series, exhibiting elevated error rates across all regression metrics with RMSE values ranging from 0.48 to 0.769 and \$MAPE scores between 17.815% and 24.626%. This is likely attributable to their inherent characteristics: they attempt to generate new points based on the initial five and noise, rendering them more susceptible to higher error rates, especially in a multi-ticker context.

In addition, the impact of sentiment integration varies significantly across architectural implementations, revealing complex interactions between NLP capabilities and temporal modeling techniques. A closer examination shows that the LSTM architecture effectively integrates sentiment information, achieving the best performance with the Vicuna model marked by a lower loss compared to the economic baseline. In contrast, the Llama and Mistral variants display slight performance degradation, suggesting that LSTM models possess selective sensitivity to the sentiment quality derived from an LLM.

The GAN model exhibits the most intricate sentiment integration landscape with Vicuna attaining the optimal generator loss and \$MAPE, indicating that adversarial training can exploit high-quality sentiment features effectively. Nonetheless, all sentiment-augmented GAN versions underperform their economic-only baselines in RMSE. This underscores the inherent challenge of effectively incorporating sentiment data within adversarial training frameworks for financial forecasting.

Transformer models demonstrate competitive performance across multiple metrics with Vicuna also achieving strong overall results in this case, which suggests that transformer architectures may possess inherent capabilities for processing traditional financial indicators that partially obviate the need for explicit sentiment features while still benefiting from sentiment integration.

In Fig. 2, we present violin plots for Vicuna, the best-performing LLM for polarity sentiment extraction, to examine the percentage variation errors across the three architectures, focusing on their distribution as well as mean and median values.

The LSTM model shows superior error concentration with a mean error of 5.63% and a median of 2.99% (mean-to-median ratio of 1.88), which highlights a reasonably narrow error distribution and low impact from outliers, supporting its reliability and consistency in financial forecasting applications where consistent performance is crucial.

In comparison, the transformer model shows a slightly higher error dispersion with a median of just 1.08% and a mean of 5.71% (mean-to-median ratio of 5.29) which underlines significant contrast and frequent outlier predictions, which lead to a decrease in the model's overall predictive consistency, despite high best-case performance.

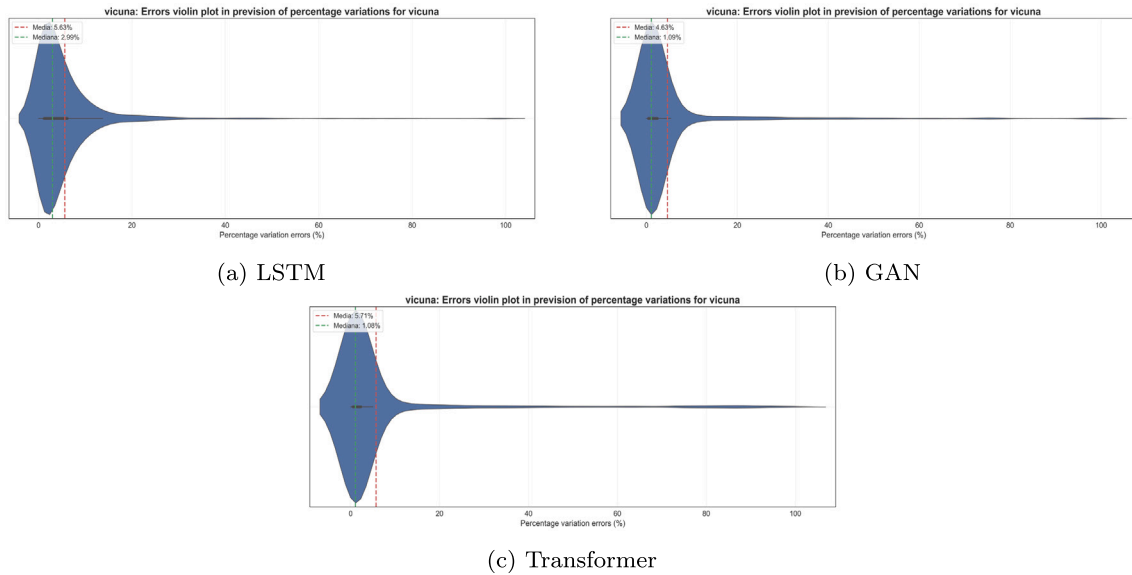


Fig. 2. Error distribution of the three prediction models through violin plots.

Table 10

Computational efficiency. It is worth noting that “it” and “tt” acronyms represent the inference and training time, respectively.

Model	$LSTM_{it}$	$LSTM_{tt}$	GAN_{it}	GAN_{tt}	$Transformer_{it}$	$Transformer_{tt}$
Llama	3.6560	943.0320	4.0270	6166.6820	0.3743	972.7260
Vicuna	4.4800	958.0340	5.4260	6089.2120	0.3271	989.2290
Mistral	5.7660	957.6490	6.5410	6130.3780	0.3354	981.1140
Economic	6.9050	957.8520	7.1370	6189.1230	0.3289	1016.1540

The GAN model has a comparable skewed error profile exhibiting a median of 1.09% and a mean of 4.63% (mean-to-median ratio of 4.25), which suggests a heavy-tailed distribution. Moreover, while its best-case predictions are competitive, the huge influence of outliers limits its dependability in real-world financial contexts.

Mean-to-median ratio analysis highlights the stability discrepancies between models, revealing that LSTM has the lowest ratio, addressing superior robustness and error control. The Transformer and GAN models on the counterpart show higher ratios, a factor that is indicative of increased unpredictability.

In Table 10, we examine the computational efficiency of the models. It is possible to highlight substantial disparities in training requirements, which have considerable ramifications for practical implementation and research scalability. LSTM models exhibit the best computational efficiency with training times ranging from 943 to 958 seconds, reflecting six to seven times faster convergence compared to GAN designs. On the other hand, transformer models hold an intermediate position, indicating around 15% longer training time elapsed than LSTM implementations while preserving major efficiency advantages over GAN techniques.

In Fig. 3, we show the closing price predictions related to AMD⁶ ticker, which is one of the tickers with more data. We identify significant differences in trend tracking, volatility response, and alignment with actual market behavior.

A global analysis demonstrates that the Vicuna-based prompts yield more accurate and stable predictions, especially when used in conjunction with transformer architectures. Further visual analysis of forecast trajectories reveals important differences in model behavior that are not fully captured by \$MAPE alone, because it is evident that transformer outputs tend to exhibit smoother curves and more coherent alignment with actual market trends. These models were observed to follow turning points in the price series with relatively low lag and minimal erratic jumps. GANs, while occasionally achieving close numerical predictions, often produced visually unstable forecasts characterized by sharp discontinuities and implausible directional changes. The LSTM model showed smoother but sluggish behavior, often lagging behind actual price inflections and failing to capture short-term dynamics. The distribution of prediction errors can explain the difference between \$MAPE results and visual coherence. Indeed, GANs and Transformers models both produce many small errors, characterized by a lower \$MAPE than the LSTM one, but Transformers and especially LSTM tend to distribute those errors more uniformly, avoiding outliers. Conversely, GANs can achieve very good point estimates in isolation but may do so with less interpretable or unstable dynamics, which could be problematic for decision-making contexts like algorithmic

⁶ <https://www.amd.com/en.html>.

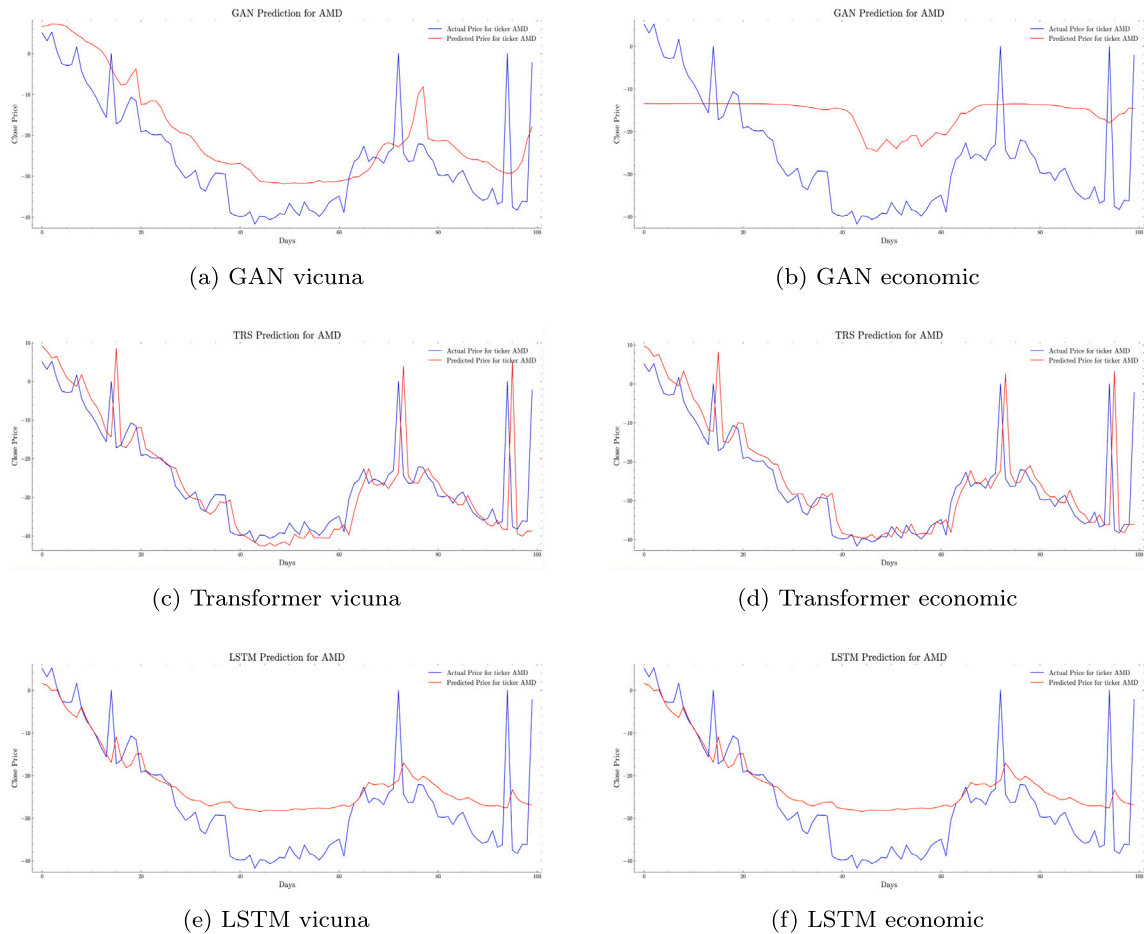


Fig. 3. AMD stock prediction using Vicuna against the economic indicators baseline.

trading. So, as an overall comparison, both for computational efficiency and general forecasting prices, LSTM appears to provide better-balanced results.

These experimental observations align with established theoretical frameworks in financial time-series literature. We further contextualize our framework's performance by discussing the results obtained against the theoretical findings of a representative model taxonomy [18] composed of Recurrent Neural Networks, Transformers, and GANs.

Recurrent models such as LSTMs are widely recognized for their ability to model sequential dependencies and smooth temporal transitions within noisy financial environments. Their ability to maintain stable regression performance and lower variance in prediction errors, as evidenced by our lower RMSE dispersion and reduced mean-to-median error ratios [26], is well supported by recent comparative studies [30]. Similarly, our findings regarding Transformer-based architectures reflect their capacity for capturing long-range temporal dependencies through self-attention mechanisms [31]. While they proved computationally more demanding in our tests, their competitive RMSE and superior \$MAPE scores suggest an enhanced ability for local trend adaptation and short-term sensitivity. This confirms that attention-based models can effectively synthesize heterogeneous signals, such as the integration of sentiment and traditional financial indicators [30]. Conversely, GAN-based forecasting approaches have been previously associated with higher training instability due to adversarial optimization dynamics [32]. The elevated variance and larger outlier sensitivity observed in our results are theoretically coherent with the known difficulty of stabilizing GAN training in regression-oriented financial forecasting scenarios, particularly in environments characterized by heterogeneous volatility regimes [33].

Finally, we conducted an XAI evaluation to explore the underlying factors driving these substantial performance differences across the various LLMs.

6.1. XAI

To analyze the differences in model outputs, we selected nine headlines, three each, classified as neutral, positive, and negative by the Vicuna model, and compared their classifications with those provided by Llama and Mistral, as shown in Table 11 (Table 12).

The polarity prediction results demonstrate considerable disparities in how each language model interprets sentiment in financial headlines. Indeed, as depicted in the heatmap in Fig. 4, Llama gives extremely polarized scores, whereas Vicuna tends to be more

Table 11
Sentiment analysis performance Across models.

Model	Sentiment	Polarity	Category	Text Sample	Key Evidence
Vicuna					
•	Neutral	+0.60	Moderate	<i>Best Stocks Of 2021: Meme Stocks...</i>	Model assigned positive scores to...
•	Neutral	+0.50	Moderate	<i>The 25 Best-Performing Stocks...</i>	Positive sentiment based on...
•	Neutral	+0.50	Moderate	<i>10 Top Robinhood Stocks That...</i>	Financial context considered...
•	Positive	+0.95	Strong	<i>Hedge Funds Salvage 2021 Gains...</i>	Performance of hedge funds...
•	Positive	+0.95	Strong	<i>ON AB Small Cap Portfolio Buys...</i>	Portfolio buying activity...
•	Positive	+0.90	Strong	<i>3 Stocks to Buy On the Dip...</i>	Investment opportunity signal...
•	Negative	−0.80	Strong	<i>AMC Stock Could Very Well Be...</i>	Downward trend indication...
•	Negative	−0.80	Strong	<i>Why Loopring Is Up Over 620%...</i>	Contradictory model assignment...
•	Negative	−0.80	Strong	<i>GameStop Stock Looks Exceptionally...</i>	Unlikely turnaround prospects...
Llama					
•	Positive	+0.80	Strong	<i>Best Stocks Of 2021: Meme Stocks...</i>	'Best Stocks' and 'Broad...'
•	Positive	+0.80	Strong	<i>The 25 Best-Performing Stocks...</i>	Positive outlook on stocks...
•	Positive	+0.70	Moderate	<i>10 Top Robinhood Stocks That...</i>	Stock market doubling...
•	Positive	+0.70	Moderate	<i>Hedge Funds Salvage 2021 Gains...</i>	'Salvage' negative connotation...
•	Positive	+0.70	Moderate	<i>3 Stocks to Buy On the Dip...</i>	'Stocks to Buy' presence...
•	Positive	+0.94	Very Strong	<i>Why Loopring Is Up Over 620%...</i>	'Up' and '620%' indicators...
•	Negative	−0.80	Strong	<i>AMC Stock Could Very Well Be...</i>	'way down for good' phrase...
•	Negative	−0.80	Strong	<i>GameStop Stock Looks Exceptionally...</i>	'unlikely' negative connotation...
•	Neutral	+0.60	Moderate	<i>ON AB Small Cap Portfolio Buys...</i>	'Buys' suggests positive...

Table 12
Sentiment Analysis Performance Across Models.

Model	Sentiment	Polarity	Category	Text Sample	Key Evidence
Mistral					
•	Neutral	+0.50	Moderate	<i>Best Stocks Of 2021: Meme Stocks...</i>	'Best Stocks' mentions...
•	Neutral	+0.60	Moderate	<i>10 Top Robinhood Stocks That...</i>	'Top Robinhood Stocks'...
•	Neutral	+0.60	Moderate	<i>Hedge Funds Salvage 2021 Gains...</i>	'Salvage 2021 Gains'...
•	Neutral	+0.40	Weak	<i>ON AB Small Cap Portfolio Buys...</i>	'buys' and 'positive' signals...
•	Neutral	+0.60	Moderate	<i>3 Stocks to Buy On the Dip...</i>	Investment opportunity...
•	Positive	+0.80	Strong	<i>The 25 Best-Performing Stocks...</i>	'best-performing' mention...
•	Positive	+0.80	Strong	<i>Why Loopring Is Up Over 620%...</i>	Large increase indication...
•	Negative	−0.80	Strong	<i>AMC Stock Could Very Well Be...</i>	Downward trend suggestion...
•	Negative	−0.80	Strong	<i>GameStop Stock Looks Exceptionally...</i>	Turnaround unlikely...

conservative, with few instances of neutral forecasts. Mistral exhibits moderate behavior, typically coinciding with Llama on polarity direction but with lower intensity.

For instance, headlines such as “Hedge Funds Salvage 2021 Gains...” and “Why Loopring Is Up Over 620%...” receive significantly positive polarity both from Llama and Mistral, while Vicuna diverges sharply in the latter case, assigning a polarity of −0.8, suggesting model-specific interpretive heuristics. Indeed, the gap might be explained by Vicuna’s increased sensitivity to underlying ambiguity or contextual uncertainty in terms such as “up over 620%,” which might be perceived as unsustainable or speculative rather than simply positive.

On the other hand, the boxplot in Fig. 5, comparing polarity distribution by ambiguity existence, further validates this observation. Texts marked as ambiguous (i.e., where financial implications are not explicitly positive or negative) are associated with wider variance and a lower median polarity, a factor that implies that models exhibit increased uncertainty or conservative scoring when ambiguity is present, aligning perfectly with Vicuna’s outputs, which follow this trend, consistently lowering polarity in ambiguous cases.

Analyzing the annotated XAI results, Vicuna’s reasoning process regularly refers to ambiguity and specifically indicates its consideration in several situations. For instance, Vicuna assigns neutral scores to headlines that mention general performance without directional indicators (e.g., “Best Stocks of 2021”) and justifies this based on the lack of explicit cues, differentiating itself from Llama and Mistral, which rely more heavily on surface-level textual positivity (e.g., terms like “top,” “best-performing”) and tend to ignore subtle ambiguity.

The structured XAI explanations reveal underlying decision-making processes that align with model behavior:

- Vicuna often includes financial context in its explanation pipeline. It refers to ambiguous aspects and reasoning under uncertainty and suggests a more cautious approach in uncertain headlines;
- Mistral and Llama, on the other hand, appear more reliant on textual content only with limited reference to ambiguity and their decisions. While directionally consistent, they may lack depth in disambiguating nuanced sentiment.

This divergence is confirmed by Vicuna’s reasoning, which more frequently mentions the ambiguity in the explanation, showing its explicit technique for handling uncertainty, likely contributing to its more conservative polarity distribution.



Fig. 4. Heatmap related to the polarity scores yielded by Llama, Mistral, and Vicuna LLM models shows remarkable differences based on a different reasoning process.

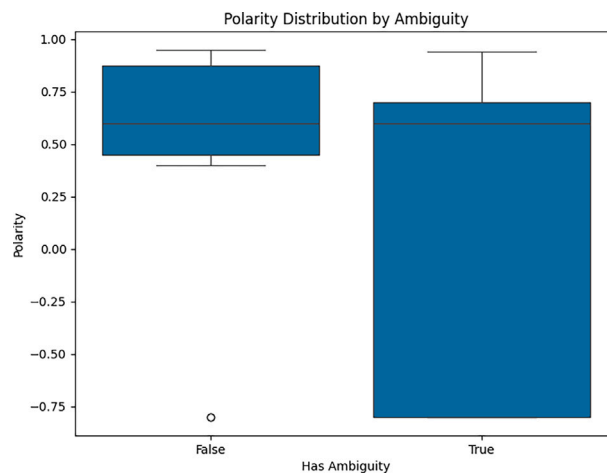


Fig. 5. Polarity distribution by ambiguity.

7. Conclusion and future work

Stock forecasting is a complex and high-impact task, particularly when integrating unstructured data sources such as financial news. In this work, we have presented a comprehensive framework that leverages LLMs for zero-shot sentiment analysis of news headlines and integrates these sentiment signals with historical stock price data. We combine this enriched dataset with deep learning models to improve stock price prediction effectiveness.

The experimental results reveal that large performance disparities exist among LLM sentiment extractors, with Vicuna emerging as the most successful model for financial sentiment integration despite exhibiting the greatest hallucination rate at 9.6% among the LLMs utilized by us. This research highlights intrinsic trade-offs between computational efficiency and output reliability that demand careful consideration for practical deployment scenarios.

By our results, we have seen that architectural selection makes an impact on both prediction accuracy and computational efficiency, with LSTM models consistently demonstrating superior performance across multiple metrics, achieving the lowest RMSE values while

requiring substantially less computational resources compared to alternative architectures. However, Transformer models achieve competitive accuracy with superior MAPE performance, even while demanding increased computational overhead, whereas GAN architectures exhibit fundamental limitations for direct financial forecasting applications despite their theoretical advantages in capturing complex data distributions.

From this investigation, we found that enhanced dataset integration represents a critical priority, requiring the development of more comprehensive datasets that incorporate several information sources beyond traditional news headlines, with the integration of social media sentiment, earnings call transcripts, regulatory filings, and real-time market commentary, to create richer multimodal datasets while investigating alternative temporal granularities and cross-market sentiment propagation for deeper insights into global market dynamics.

While our work is focused on meme stocks, recent literature suggests that long-term relationships (e.g., industry linkages) also have transient effects [4]. This indicates that even for traditional blue-chip stocks, external information (e.g., news, events) can still play a role, albeit with different temporal dynamics due to challenges in interpreting event semantics [1]. This suggests that the consistency and contextual interpretation of sentiment remain important factors. Finally, the use of recent NLP methodologies (e.g., LLMs) improves the extraction and interpretation of financial textual data, potentially mitigating some of the noise associated with lower or less reactive news volume [34].

Future improvements in forecasting performance may come from advanced architectures, such as ensembles and lightweight models augmented with attention mechanisms. These approaches could combine the efficiency of recurrent networks with the long-range pattern recognition of attention-based models.

From a deployment perspective, real-time implementation and scalability research should be addressed as the main challenges in developing streaming processing pipelines for real-time sentiment analysis and prediction. This should also take into account different techniques, such as edge computing solutions and model compression, which can help implement the solution in resource-constrained environments.

Based on dataset typology, both cross-market and multi-asset extensions could greatly enhance the practical usefulness of sentiment-augmented forecasting through studying the generalizability of findings across diverse asset classes, geographic marketplaces, and economic conditions.

CRediT authorship contribution statement

Giovanni Officioso: Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Giancarlo Sperli:** Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Andrea Vignali:** Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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